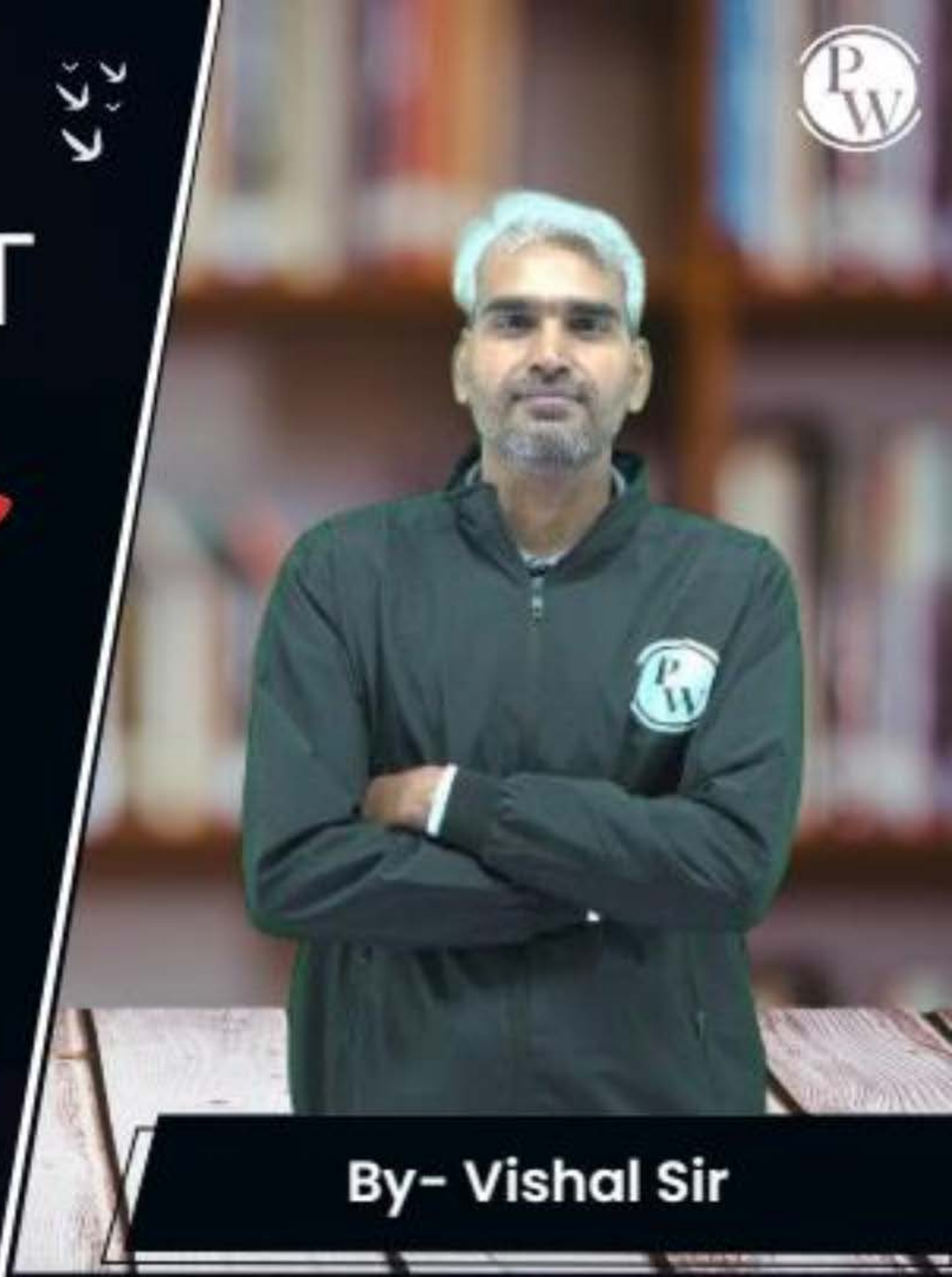
Computer Science & IT



Discrete Mathematics

Mathematical Logic

Lecture No. 05



By- Vishal Sir

Recap of Previous Lecture



- Topic — Practice questions
- Topic — Proof by contradiction
- Topic — Conditional proof
- Topic — Predicate logic

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Topics to be Covered



- Topic: Quantifiers
- Topic: Scope of quanitifiers
- Topic: Negation of statement formula
- Topic: Sentences with multiple quantifiers
- Topic: Relationship diagram

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Predicate Logic

(First order predicate logic)



Topic : Predicate

It is part of a sentence or clause stating something about the subject.

eg: Ram is a Sportsman

Ram — Subject

is a Sportsman — Predicate

] it is a proposition
it may be true
or
it may be false

- Let 'S' denotes "is a Sportsman"
  Here 'S' is a predicate.

S(Ram) : Ram is a Sportsman

Predicate S will be applied on subject Ram

P: is a politician

S: is a sportsman

- P(x): x is a politician
- S(y): y is a sportsman

} x & y may be any person from the set of people under consideration

↑ it is our domain

$P(x) \rightarrow S(x)$: if P(x) then S(x)

: if x is a politician, then x is a sportsman

it may be true or it may be false

① if 'x' is not a politician, then implication will return true, does not matter whether x is a sportsman or not.

② If x is a politician then x must be sportsman as well for implication to return true.

③ If x is a politician but not a sportsman, then implication will return false

P: is a politician

S: is a sportsman

① $P(x) \rightarrow S(y)$: if x is a politician, then y is a sportsman

② $P(x) \vee S(y)$: x is a politician or y is a sportsman

③ $P(x) \wedge S(x)$: x is politician cum sportsman

④ $P(x) \leftrightarrow S(y)$: x is a politician if and only if y is a sportsman

if 'p' : is a Politician.

then '~p' : is not a Politician

P(Ram) ≡ Ram is a Politician.

~P(Ram) ≡ Ram is not a Politician.



Topic : Predicate with multiple subjects

Let 'F' : is a friend of

F(x,y) : x is a friend of y

- '~F' : is not a friend of

~F (Himanshu, Krishna) : Himanshu is not a friend of Krishna

Let G: is greater than

G(a, b): a is grater than b

{May be true or may be false}

eg: G(2, 3): is false

G(5, 5): is false

G(5, 3): True

e.g: $F(x, y, t)$: Person 'x' can fool person 'y' at time 't'

$F(\text{Vishal}, \text{Himanshu}, 11{:}00\,\text{PM})$

Today's Topic

$$S(\text{Sumit}) \vee S(\text{Niranjan}) \vee S(\text{Hariom}) \vee - - - - \vee - .$$

$$S(\text{Sumit}) \wedge S(\text{Niranjan}) \wedge S(\text{Hariom}) \wedge - - - - \wedge - .$$



Topic : Quantifiers

- In predicate logic, predicates are used alongside quantifiers to express the extent to which a predicate is true over a range of elements.
  - i.e, over the elements of domain or over the elements of Universe of discourse
- There are two types of quantifiers

1. Universal Quantifier — $\forall$: for all
2. Existential Quantifier — $\exists$: there exists (for some / for at least one)



Topic : Universal Quantifier (∀)

- $\forall$: for all
- $\forall_x$: for all 'x' in the domain
  or
  for all 'x' in the universe of discourse

Let S : is a sportsman
& domain is the set of all students in our class

$\forall_x \{ S(x) \}$: All are sportsman { This statement is trying to say that all students in our class are sportsman }

It may be true
or It may be false

$\forall_x \{ S(x) \}$ → Truth value of this predicate formula is the truth value returned by '∀'

- $\forall x \{S(x)\}$ will return false if and only if there is at least one element in the universe of discourse {i.e: in the domain} for which the Predicate S(x) returns false

Note: If universe of discourse (i.e: domain) happens to be Empty, then quatifier "$\forall$" will always return true, because there is no element for which the underlying Predicate can return false

* $\forall x \{S(x)\}$ will return true if and only if,

① domain is Empty

(or) ② Predicate 'S' returns true for all the elements in the domain

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In $\forall x \{S(x)\}$

$\forall x$: quantifier

$\{S(x)\}$: Scope of the quantifier

$\forall x \{P(x) \rightarrow S(x)\}$

$\{P(x) \rightarrow S(x)\}$: Scope of quantified

Q:: Which of the following Predicate Formula is/are valid

F_1: $\left[\forall x \{P(x)\} \longrightarrow \forall x \{S(x)\}\right] \longrightarrow \left[\forall x \{P(x) \longrightarrow S(x)\}\right]$

F_2: $\left[\forall x \{P(x) \longrightarrow S(x)\}\right] \longrightarrow \left[\forall x \{P(x)\} \longrightarrow \forall x \{S(x)\}\right]$

(a) F_1 only (b) F_2 only

(c) Both F_1 & F_2 (d) Neither F_1 nor F_2

Let, Domain is set of all living things

& Let $H(x)$: x is a human

& $I(x)$: x is intelligent.

What is represented by following formulea

① $\forall x \{H(x)\} \longrightarrow \forall x \{I(x)\} \equiv$ if $\forall x \{H(x)\}$ then $\forall x \{I(x)\}$

$\therefore$ if all are human then all are intelligent

② $\forall x \{H(x) \longrightarrow I(x)\}$: for all x, if $H(x)$ then $I(x)$

$\equiv$ for all x, if x is human then x is intelligent

$\equiv$ All human are intelligent

for it to be true,
either x is not human or
if x is human then x must be intelligent

Q:: Which of the following Predicate Formula is/are valid

F_1: $\left[\forall x \{P(x)\} \longrightarrow \forall x \{S(x)\}\right] \longrightarrow \left[\forall x \{P(x) \longrightarrow S(x)\}\right]$

F_2: $\left[\forall x \{P(x) \longrightarrow S(x)\}\right] \longrightarrow \left[\forall x \{P(x)\} \longrightarrow \forall x \{S(x)\}\right]$

(a) F_1 only (b) F_2 only

(c) Both F_1 & F_2 (d) Neither F_1 nor F_2

Let P: is a politician S: is a sportsman

$\forall x\{P(x) \rightarrow S(x)\}$: for all x '$P(x) \rightarrow S(x)$' is true

We are checking whether for all 'x' in the domain $P(x) \rightarrow S(x)$ is true or not

* Truth value of this predicate formula is the truth value returned by '$\forall$'

* If there exists any 'x' for which P(x) is true but S(x) is false, then '$P(x) \rightarrow S(x)$' will be false and hence '$\forall$' will return false
- Predicate formula will be true only when $P(x) \rightarrow S(x)$ is true for all { whenever P(x) is false, then $P(x) \rightarrow S(x)$ will always be true, But whenever P(x) is true then S(x) should also be true for that x

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$\forall x\{P(x)\} \rightarrow \forall x\{S(x)\}$: if $\forall x\{P(x)\}$ then $\forall x\{S(x)\}$

LHS R.H.S

i.e, if all are politician then all must be sportsman

Truth value of this predicate formula is the truth value returned by implication

* If there is any person in the domain who is not a politician, then $\forall x\{P(x)\}$ will be false, i.e, L.H.S. of implication will be false ∴ implication will return true
* But if all are politician in the domain, then L.H.S will be true ∴ for implication to be true, all must be sportsman aswell

F_1: $[\forall x \{P(x)\} \longrightarrow \forall x \{S(x)\}] \longrightarrow [\forall x \{P(x) \longrightarrow S(x)\}]$

P → Q

$U = \{a, b, c\}$

$P(a) = T$
$P(b) = F$
$P(c) = T$

$\therefore \forall x \{P(x)\} = \text{False}$

in this case
if $S(a) = F$
$S(b) = T$
$S(c) = T$

then

$P(a) \rightarrow S(a) \equiv T \rightarrow F = F$
$P(b) \rightarrow S(b) \equiv F \rightarrow T = T$
$P(c) \rightarrow S(c) \equiv T \rightarrow T = T$

$\therefore \forall x \{P(x) \rightarrow S(x)\}$ will return false

$\therefore \forall x \{P(x)\} \rightarrow \forall x \{S(x)\}$ is true because LHS is false

When LHS is True, then RHS may be false ∴ invalid.

F_2: $[\forall x \{ P(x) \longrightarrow S(x) \}]$ $\longrightarrow$ $[\forall x \{P(x)\} \longrightarrow \forall x \{S(x)\}]$

if $P(x)$ is false for at least one x then $\forall x\{P(x)\}$ is false, i.e. $\forall x\{P(x)\} \rightarrow \forall x\{S(x)\}$ is definitely true, i.e. R.H.S of implication will be true ∴ whole implication will be true

for it to be true whenever 'P' is true then 'S' must be true

But Let R.H.S = False

i.e. let $\forall x \{P(x)\}$ = True & $\forall x\{S(x)\}$ = False

↓

i.e. S is false for at least one x

if P(x) is true for all, but S(x) is false for at least one 'x'

Hence $P(x) \rightarrow S(x)$ is definitely false for at least one x

∴ $\forall x \{P(x) \rightarrow S(x)\}$ will return false i.e. L.H.S. will also be false

When RHS is false, then LHS is guaranteed to be false, Hence implication is valid

Q:- Which of the following Predicate Formula is/are valid

F_1: $[\forall x \{P(x)\} \longrightarrow \forall x \{S(x)\}] \longrightarrow [\forall x \{P(x) \longrightarrow S(x)\}]$

F_2: $[\forall x \{P(x) \longrightarrow S(x)\}] \longrightarrow [\forall x \{P(x)\} \longrightarrow \forall x \{S(x)\}]$

(a) F_1 only (b) F_2 only

(c) Both F_1 & F_2 (d) Neither F_1 nor F_2

H.W: Consider following two Predicate formula

F_1 : $\forall x \{P(x) \longleftrightarrow Q(x)\} \longrightarrow \{\forall x \{P(x)\} \longleftrightarrow \forall x \{Q(x)\}\}$

F_2 : $\{\forall x \{P(x)\} \longleftrightarrow \forall x \{Q(x)\}\} \longrightarrow \forall x \{P(x) \longleftrightarrow Q(x)\}$

Which of the following is true

(a) Only F_1 is valid
(b) Only F_2 is valid
(c) Both F_1 and F_2 are valid
(d) Neither F_1 nor F_2 is valid



Topic : Existential Quantifier (∃)

$\exists$: there exists
for some / for at least one

$\exists_x$: for some 'x' in the domain
for at least one (or) x in the domain

let P : is a Politician
and domain is set of all students in our class.

then $\exists_x \{P(x)\}$: for at least one 'x', x is a politician
(for some)

- i.e, at least one student in our class is a politician } it may be true / it may be false

$\exists$ will return true if and only if Predicate under the scope is true for at least one element in the domain

∴ if domain is Empty then
$\exists$ will always return false

$\exists x \{ P(x) \}$

quantifier

Scope of quantifier

$\exists x \{ P(x) \rightarrow S(x) \}$

quantifier

Scope of quantifier



Topic : Scope of the Quantifier

- The part of the logical expression to which a quantifier can be applied is called the scope of the quantifier.

- A variable whose occurrence is bounded by a quantifier is called a bounded variable. Variables not bounded by any quantifiers are called free variables.

* Generally we represent the scope of the quantifier using brackets or Comma otherwise, The Scope of a quantifier is the shortest full sentence/predicate formula which follows it. Everything inside this shortest full sentence is said to be in the scope of the quantifier.



Topic : Scope of the Quantifier

- $(\forall y\ H(y) \rightarrow \exists z\ W(z, y)) \rightarrow \exists z\ G(z)$
- $\exists y(\forall y\ H(y) \rightarrow \exists z\ W(z, y)) \rightarrow \exists z\ G(z)$
- $(\forall y\ H(y) \rightarrow \exists z\ G(z)) \rightarrow \exists z\ W(z, y)$
- $(\forall z\ H(y) \rightarrow \exists y\ W(z, y)) \rightarrow \exists z\ G(z)$
- $(\forall y\ H(y) \rightarrow \exists z \exists y\ W(z, y)) \rightarrow \exists z\ G(z)$
- $(\forall y\ H(y) \rightarrow \exists z\ W(z, y)) \rightarrow \exists x\ G(z)$



Topic : Scope of the Quantifier

- $(\forall y\, H(y) \rightarrow \exists z\, W(z, y)) \rightarrow \exists z\, G(z)$

bounded variable

bounded variable

free variable

bounded

Topic : Scope of the Quantifier

- $\exists y(\forall y\, H(y) \rightarrow \exists z\, W(z, y)) \rightarrow \exists z\, G(z)$

bounded by $\forall y$

bounded

bounded

bounded

All free occurance of 'y' in the scope of $\exists$ will be quantified by $\exists$

Same Predicate formula can also be written in the following format

$$\exists x(\forall y\, H(y) \rightarrow \exists z\, W(z, x)) \rightarrow \exists z\, G(z)$$

A single variable can not be quantified using multiple quantifiers

↓

Even if it seems so then also variable will be quantified using inner most quantifier



Topic : Scope of the Quantifier

- $(\forall y\, H(y) \rightarrow \exists z\, G(z)) \rightarrow \exists z\, W(z, y)$

bounded

bounded

bounded

free



Topic : Scope of the Quantifier

- $(\forall z\, H(y) \rightarrow \exists y\, W(z, y)) \rightarrow \exists z\, G(z)$

free

free

bounded

bounded



Topic : Scope of the Quantifier

- $(\forall y\, H(y) \rightarrow \exists z \exists y\, W(z, y)) \rightarrow \exists z\, G(z)$

bounded

bounded

bounded

bounded



Topic : Scope of the Quantifier

- $(\forall y\, H(y) \rightarrow \exists z\, W(z, y)) \rightarrow \exists x\, G(z)$

bounded bounded free free



2 mins Summary

Topic Quantifiers

Topic Scope of quantifiers

Topic Negation of statement formula

Topic Sentences with multiple quantifiers

Topic Relationship diagram

Slide


THANK - YOU
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